OPALX BeamBeam Sandbox Regression Overview

Accepted baseline: `sandbox/regression/sandbox_regression_baseline.json`
Latest metrics: `sandbox/regression/current_metrics.csv`

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This overview summarizes the sandbox regression checks. The accepted baseline values are stored in `sandbox/regression/sandbox_regression_baseline.json`; this is the comparison reference. The latest run is stored separately in `sandbox/regression/current_metrics.csv`; it is regenerated by the harness and is not the source of truth. The accepted baseline records its repository state in `_metadata.git`, shown below.

What Is Compared

Check	Input / source	Comparison target
BeamBeam field dumps	`sandbox/BeamBeam-static-1V.in`	ASCII rho/phi/E dumps in data/sandbox are compared against the analytic symmetric Gaussian manufactured solution.
Gamma-gamma e-/e+ tracking	`sandbox/track-e-p/gamma_gamma_pairs-2.in`	FROMFILE pair kinematics and final witness-container .stat summaries are checked for the primary beam, electrons, and positrons.
OPALX-IMPACT drift	`sandbox/OPALX-IMPACT/drift-1.in`	A simple proton drift with space charge is compared with digitized IMPACT reference data in extracted_graph_data.csv.

Accepted Baseline Provenance

The baseline values were accepted from the git state below. If `dirty=True`, the commit alone is not sufficient to reproduce the baseline; the listed `git status --short` entries are part of the provenance.

Item	Meaning	Value
branch	baseline source	271-implement-interation-point-element
commit	baseline source	`6ca2425c6eb2fdfe6591eb4e05bdad8a36fcbc0c`
dirty	baseline source	True

status	git status –short when accepted	`M sandbox/BeamBeam-1.in`
		`M sandbox/BeamBeam-2.in`
		`M sandbox/BeamBeam-static-1V.in`
		`M sandbox/OPALX-IMPACT/drift-1.in`
		`M sandbox/README.md`
		`M sandbox/track-e-p/analyze_pair_momenta.py`
		`M`
		`sandbox/track-e-p/convert_fort98_to_fromfile.py`
		`D sandbox/track-e-p/e-e+.gpl`
		`M sandbox/track-e-p/gamma_gamma_pairs-2.in`
		`D sandbox/track-e-p/gamma_gamma_pairs.in`
		`?? sandbox/regression/`
		`?? sandbox/track-e-p/attic/`
		`??`
		`sandbox/track-e-p/gamma_gamma_electrons-t.fromfile`
		`??`
		`sandbox/track-e-p/gamma_gamma_positrons-t.fromfile`

Physics Context From The Note

The existing note `sandbox/note/boosted_gaussian_witness.tex` defines the manufactured Gaussian witness model. The overview reuses its generated figures as visual context; the note files themselves are not part of the regression baseline.

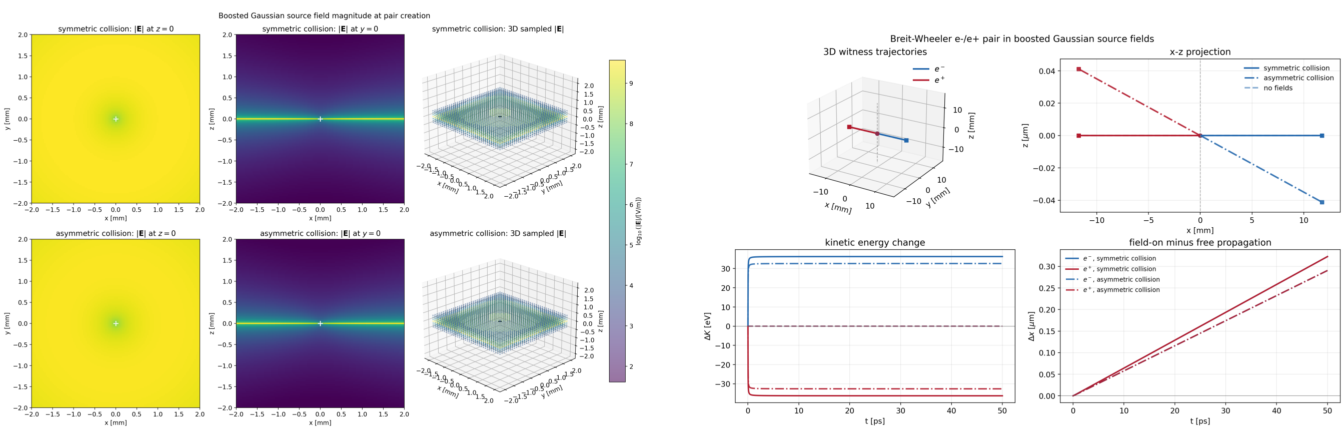

Figure 1: Field and trajectory context generated by the BeamBeam witness note.

Current Regression Metrics

BeamBeam Field Manufactured Solution

Metric	Meaning	Current value
field snapshot	diagnostic kind	active_beambeam_field_diagnostics
field centers	number of source centers	2.0
charge integral	deposited charge	-1.112650055e-14
rho relative L2	density error	0.05221858580825394
phi relative L2	potential error	0.0030422483971396385
Ex relative L2	electric field error	0.015030244723860295
Ey relative L2	electric field error	0.01582186867410125
Ez relative L2	electric field error	0.02427368611764224

OPALX-IMPACT Drift

Metric	Meaning	Current value
OPALX rows	stat samples	73.0
IMPACT rows	digitized samples	71.0
final x	OPALX final centroid	0.03478512908471119
final epsx	OPALX final emittance	1.892710694410985
RMS x difference	OPALX - IMPACT	0.0007656174663305566
max x difference	OPALX - IMPACT	0.002221940257458685
RMS epsx difference	OPALX - IMPACT	0.08641334104906384
max epsx difference	OPALX - IMPACT	0.13101914854478158

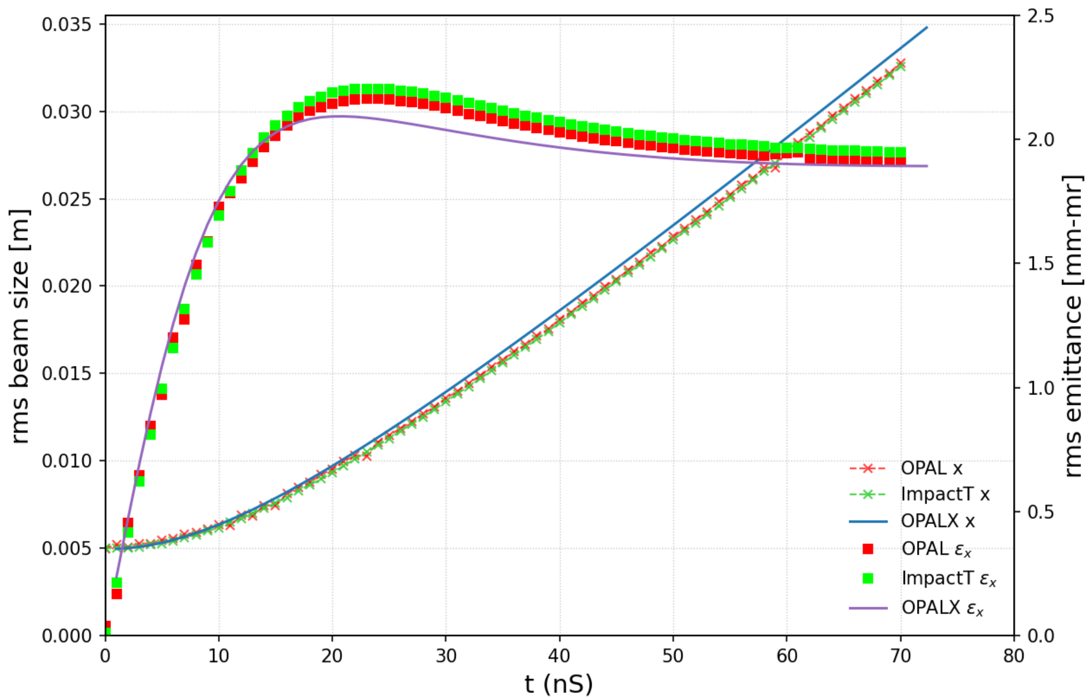

Figure 2: Digitized IMPACT reference data overlaid with the current OPALX drift result from `sandbox/OPALX-IMPACT/drift-1.stat`.

Gamma-Gamma Pair Tracking

Metric	Meaning	Current value
electron count	FROMFILE particles	1297.0
positron count	FROMFILE particles	1297.0
electron mean K	input pair kinematics	313.6014804946139
positron mean K	input pair kinematics	313.61464674324816
median opening angle	pair summary	168.90710440687806
opening > 150 deg	pair summary fraction	0.7548188126445644
electron final energy	gamma_gamma_pairs-2 c1	0.3160792683915825
positron final energy	gamma_gamma_pairs-2 c2	0.3111313465776389
electron final rms x	gamma_gamma_pairs-2 c1	0.003397430576040133
positron final rms x	gamma_gamma_pairs-2 c2	0.00338457149276078

How To Reproduce

```
source .venv-h6/bin/activate
python sandbox/regression/run_sandbox_regressions.py \
  --run-opalx \
  --opalx-exe build_openmp/src/opalx
python sandbox/regression/make_overview_pdf.py
```